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Population Projections

Lecture 3 (cont.)
Matrix Projections: Stable Populations

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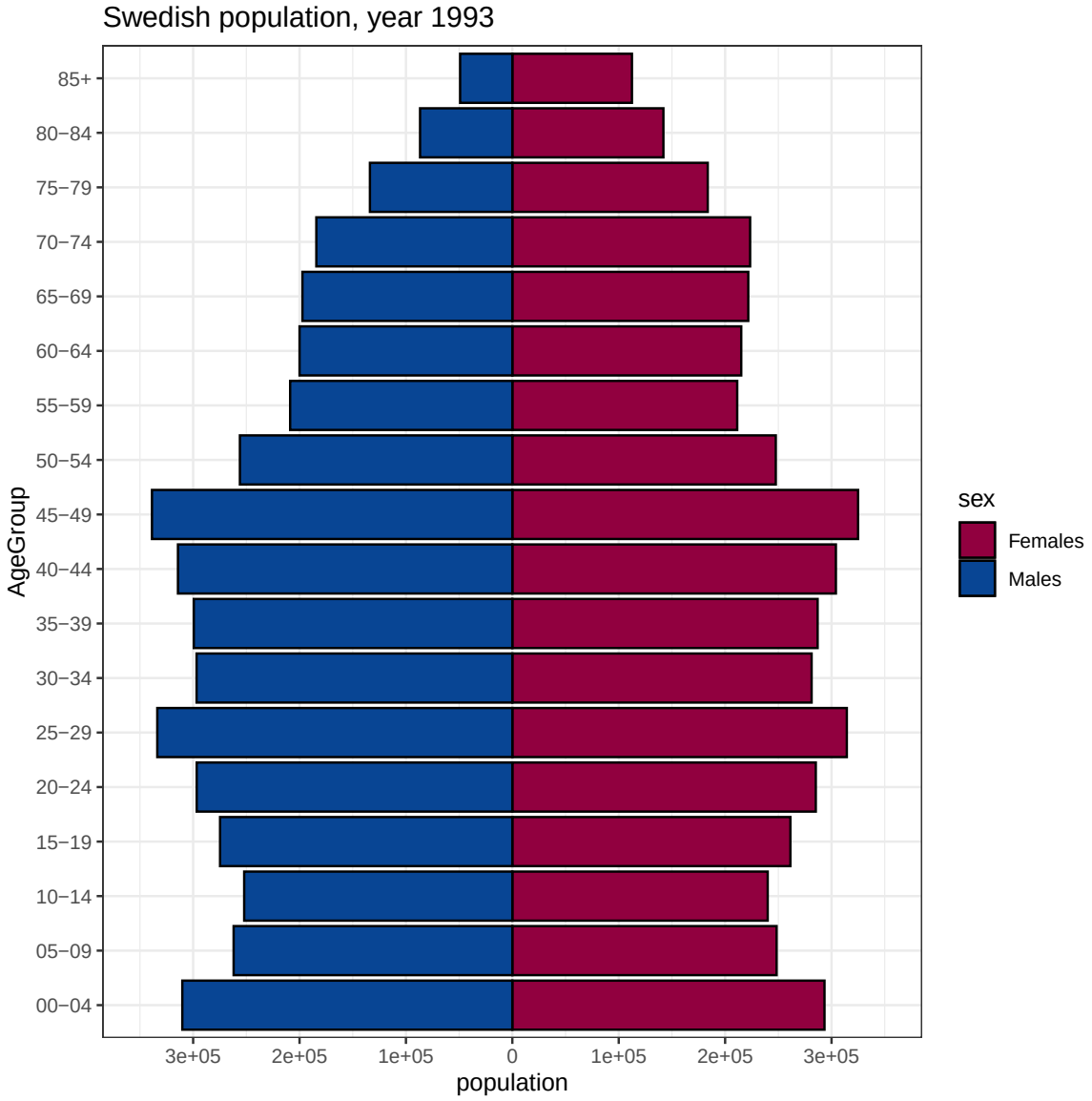
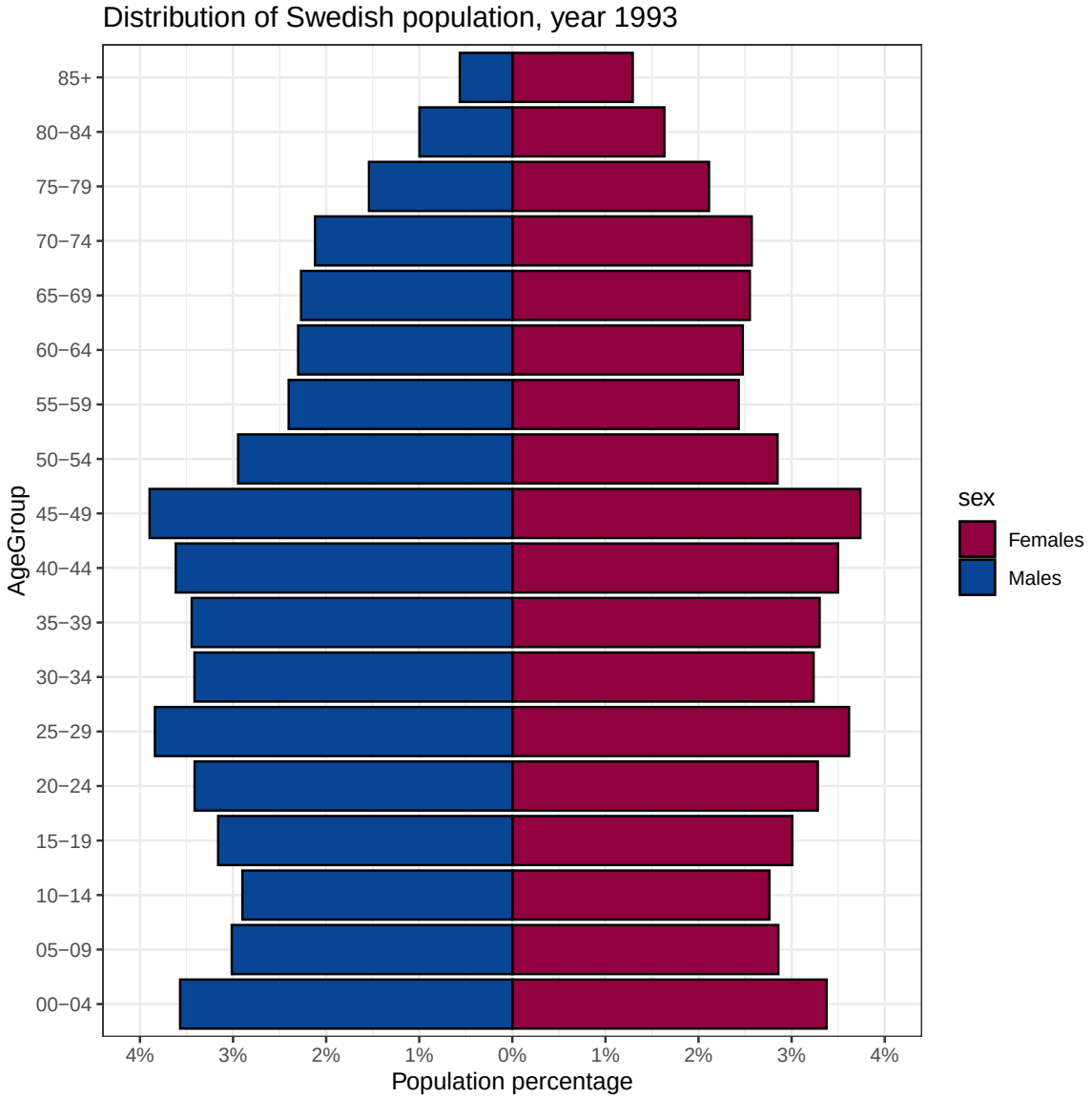
Longer projections

Two-sex closed population, constant mortality and fertility, longer projections



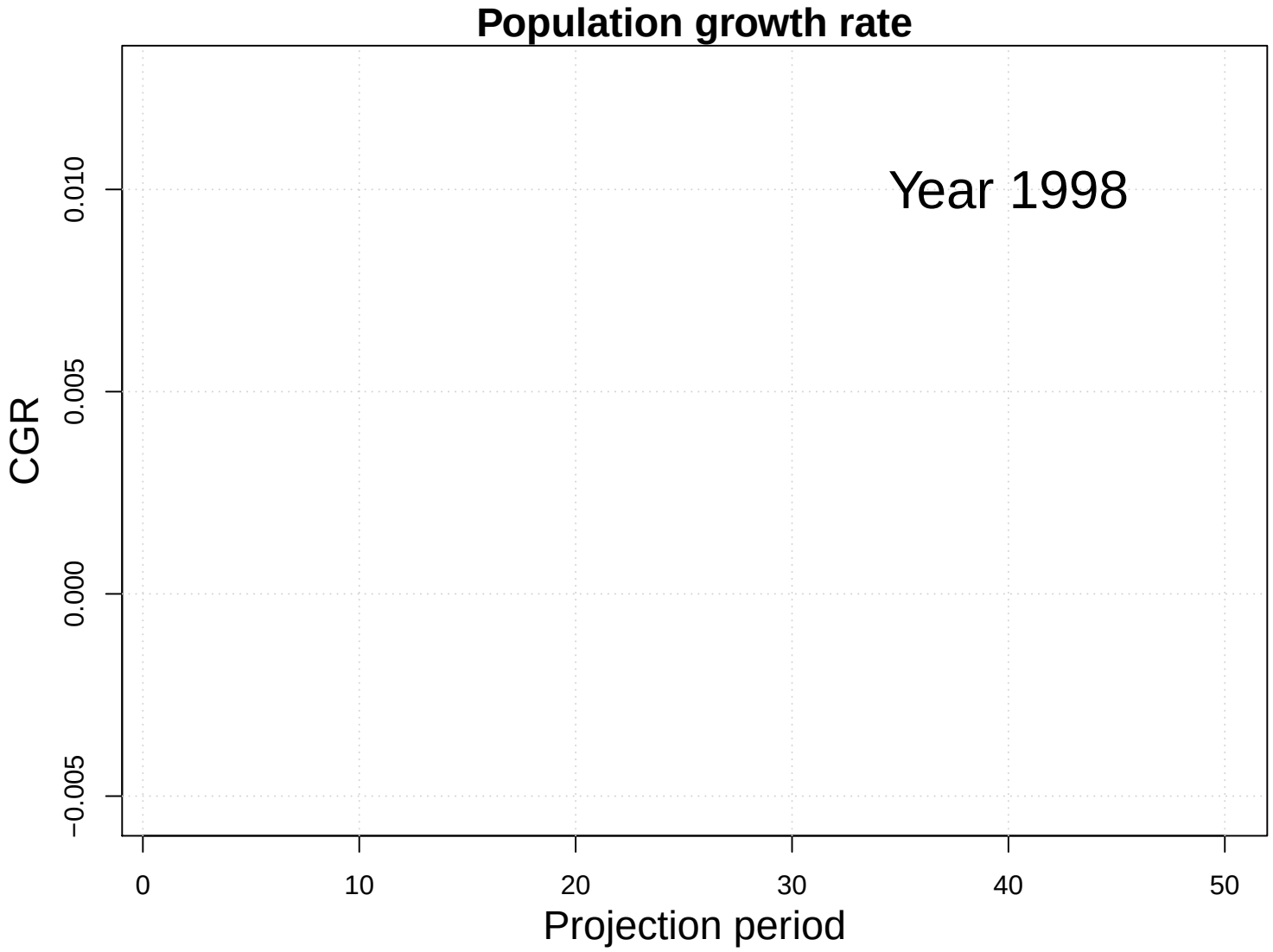
Longer projections

Let's look at the age-distribution of the population (in % terms)



Longer projections

And at the growth rate of the population



Matrix projections & stable populations

A remarkable finding about the Leslie matrix $\boldsymbol{L}$:

- under some assumptions (nearly always satisfied by human populations), when $\boldsymbol{L}$ is raised to a high enough power n, the *distribution* of the population age structure of $\boldsymbol{N}(t + 5 \times n)$ and the *crude growth rate* during each projection become **constant**
- result related to the stable population theorem

In the stable state, the population should satisfy:

$$\boldsymbol{N}^{\boldsymbol{S}}(t+5) = \boldsymbol{L}\boldsymbol{N}^{\boldsymbol{S}}(t) = \lambda \boldsymbol{N}^{\boldsymbol{S}}(t) \tag{1}$$

where λ is the crude growth rate of the population

$$\Rightarrow \text{ eigendecomposition of } \boldsymbol{L}$$

Matrix projections & stable populations

Re-arranging Equation (1)

$$\left(\boldsymbol{L} - \lambda \mathbf{I}\right) \boldsymbol{N}^{\boldsymbol{S}}(t) = 0 \tag{2}$$

where $\mathbf{I}$ is the identity matrix.

As such, the eigendecomposition of the matrix $\boldsymbol{L}$ will directly provide us with:

- the crude growth rate in the stable state, λ, which is given by the (log of the) largest real eigenvalue of $\boldsymbol{L}$
- the stable population distribution, which is given by the corresponding (rescaled) eigenvector
- refer to Caswell (2001) for additional materials on matrix population models

Matrix projections & stable populations - example

Example

```
## longer time horizon
n <- 50
my.proj <- pop.proj.v2(x=dta.swe$Age,AgeGroup=dta.swe$AgeGroup,NFx=NFx,sFx=sFx,bFx=bFx,
    NMx=NMx,sMx=sMx,bMx=bMx,n=n)

## long data
dta.swe.l <- my.proj %>%
    pivot_longer(-c(x,AgeGroup,sex),names_to = "period",values_to = "population") %>%
    mutate(period=as.numeric(period),
        Year=1993 + (period-1)*5,
        YearF=as.factor(Year))

## compute total population in each year
tot.pop <- dta.swe.l %>%
    group_by(YearF) %>%
    summarise(TotPop=sum(population))

## compute distribution of the population in each age group
dta.swe.l <- dta.swe.l %>%
    left_join(tot.pop) %>%
    mutate(Pch=population/TotPop)

## extract distribution in last year
x <- dta.swe$Age
yF <- dta.swe.l$Pch[dta.swe.l$Year == max(dta.swe.l$Year) & dta.swe.l$sex == "Females"]
yM <- dta.swe.l$Pch[dta.swe.l$Year == max(dta.swe.l$Year) & dta.swe.l$sex == "Males"]
```

Matrix projections & stable populations - example

Example

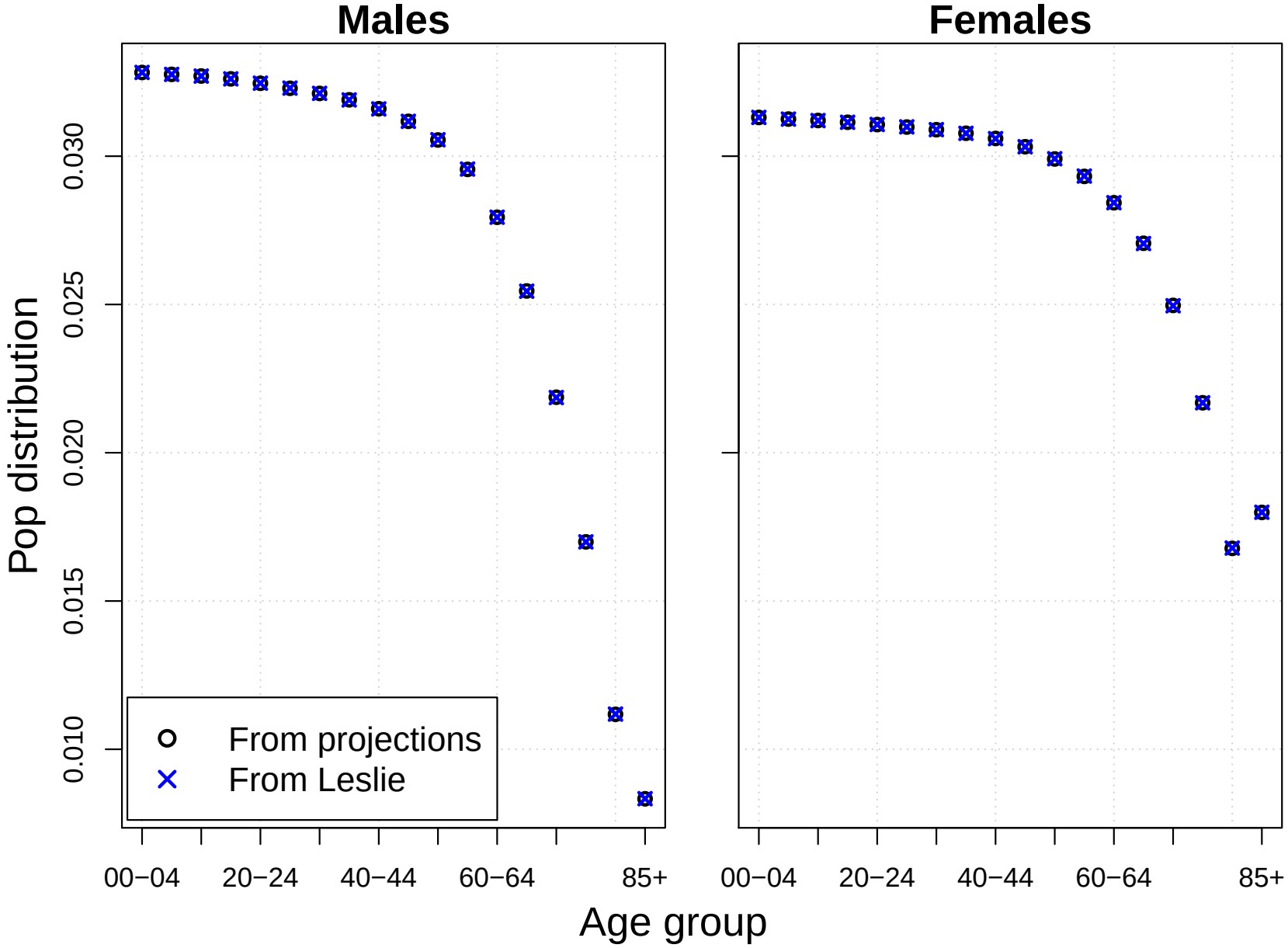
```
## eigendecomposition of the Leslie matrix
ev.decomp <- eigen(L)

## extract largest eigenvalue
## this will give you the long-term CGR of the stable population
ev.val <- abs(ev.decomp$values)[1]
cgr_leslie <- log(ev.val)

## extract corresponding eigenvectors
ev.vec <- ev.decomp$vectors[,1]
ev.vecF <- ev.vec[1:m]
ev.vecM <- ev.vec[1:m + m]

## rescale them to sum to the projected total long-term distribution
## this will give you the long-term distribution of the stable population
ev.vecF <- (ev.vecF/sum(ev.vecF)) * sum(yF)
ev.vecM <- (ev.vecM/sum(ev.vecM)) * sum(yM)
```

Matrix projections & stable populations - example



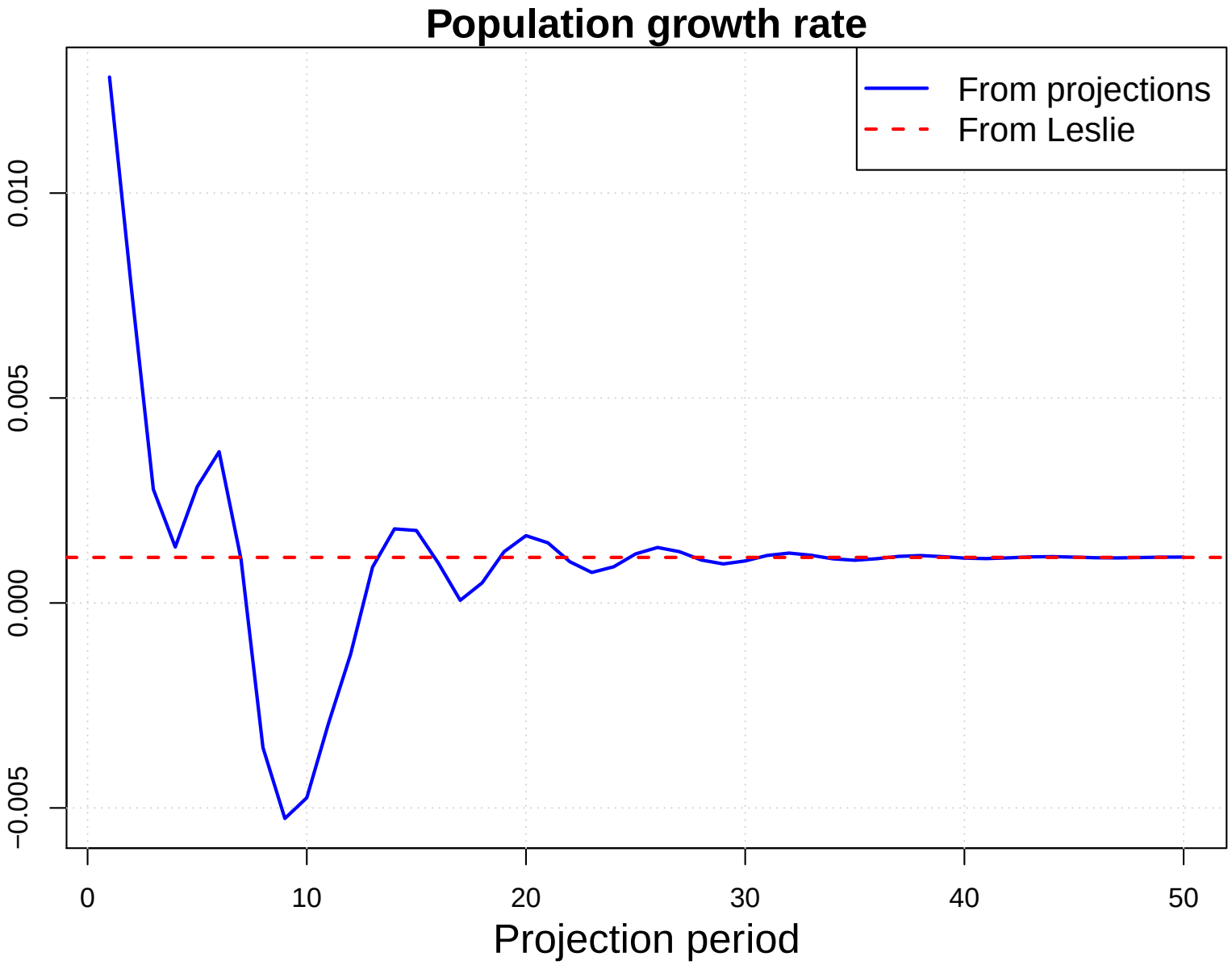
Matrix projections & stable populations - example

Example

```
## extract total population
tot.pop <- tot.pop %>%
    pull(TotPop)

## compute growth rate
CGR <- rep(NA,n)
for (i in 1:n){
    CGR[i] <- log(tot.pop[i+1]/tot.pop[i])
}
```

Matrix projections & stable populations - example



References

- Caswell, H. (2001). *Matrix Population Models*, Second Edition. Sinauer Associates, Inc. Publishers